

Project Design Phase

Solution Architecture

Date	19 June 2026	Team ID	24EG105H34
Project Name	HouseHunt – House Rent Management System	Maximum Marks	4 Marks

Architecture Description

The HouseHunt – House Rent Management System follows a **three-tier architecture** consisting of **Presentation Layer**, **Application Layer**, and **Database Layer**.

The **Presentation Layer** is developed using React.js, HTML, CSS, JavaScript, and Bootstrap to provide an interactive user interface for tenants, property owners, and administrators.

The **Application Layer** is developed using Node.js and Express.js. It handles user authentication, property management, booking management, business logic processing, and REST API services.

The **Database Layer** uses MongoDB to store user information, property details, booking records, and administrative data. MongoDB Compass is used for database monitoring and management.

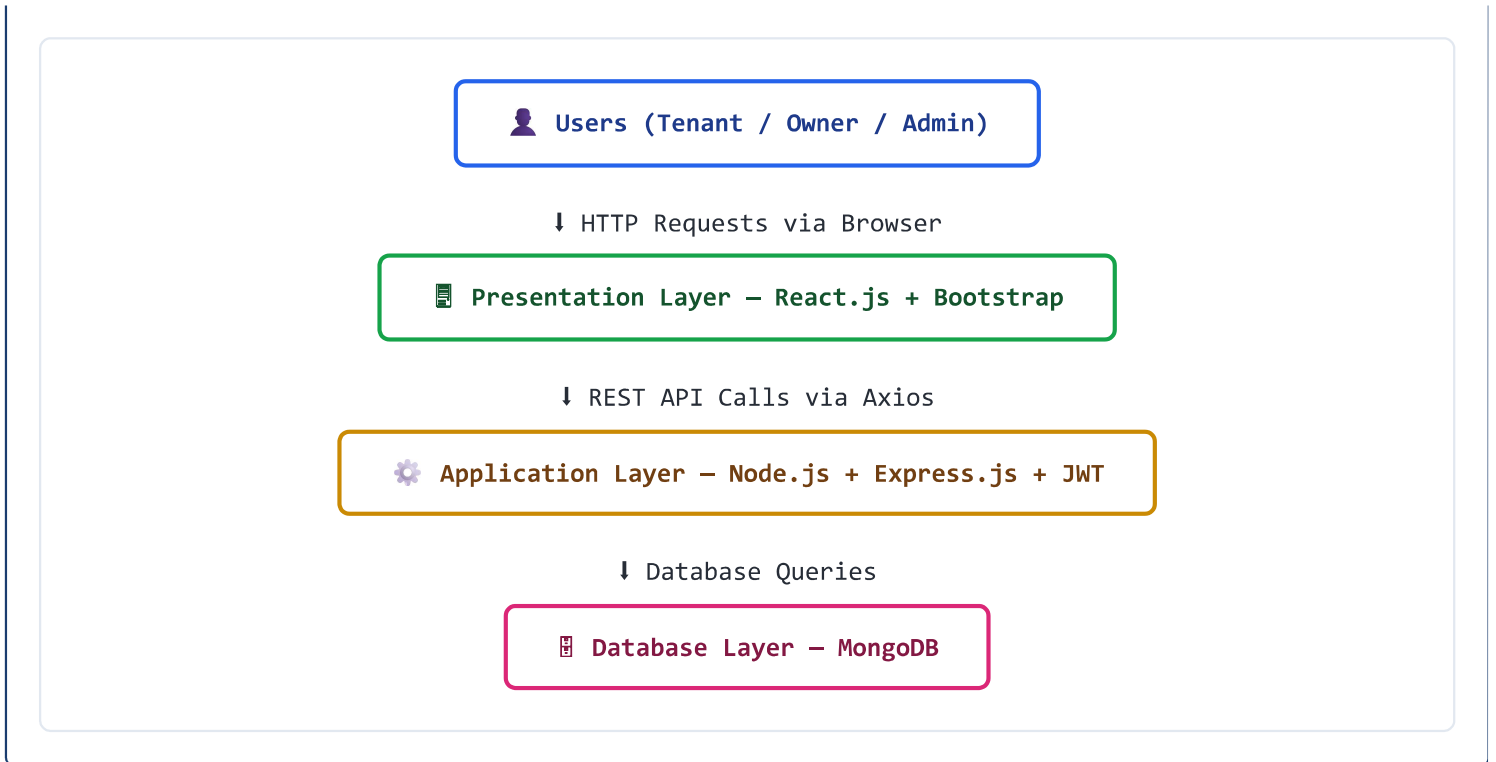
Axios is used for communication between the frontend and backend. JWT authentication and bcrypt password encryption provide secure access control and data protection throughout the system.

Components Used

Layer	Components
Presentation Layer	React.js, HTML, CSS, JavaScript, Bootstrap
Application Layer	Node.js, Express.js, REST APIs, JWT Authentication
Database Layer	MongoDB, MongoDB Compass
Communication Layer	Axios
Security Layer	JWT, bcryptjs
Development Environment	Visual Studio Code

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Solution Architecture Diagram



Data Flow

- Step 1:** User accesses the HouseHunt web application.
- Step 2:** React frontend collects user input.
- Step 3:** Axios sends HTTP requests to Express APIs.
- Step 4:** Node.js processes business logic.
- Step 5:** Express communicates with MongoDB.
- Step 6:** MongoDB stores and retrieves data.
- Step 7:** Response is returned to the frontend.
- Step 8:** Results are displayed to the user.